CITS5508 Practice Exam — Chapter 3

Classification (+ k-Nearest Neighbours)

- This practice paper has **5 questions** for **60 marks**.
- Allow approximately **two minutes per mark** (≈ 2 hours total).
- Only UWA-approved calculators permitted.
- Answers are in the separate file *Practice Exam - Chapter 3 (Answer Key).md*.

Question 1. (12 marks)

A binary classifier was evaluated, producing the following confusion matrix (rows = actual, columns = predicted):

	Predicted Negative	Predicted Positive
Actual Negative	80	20
Actual Positive	10	40

(a) Identify the number of true positives, true negatives, false positives, and false negatives. *(2 marks)*

(b) Compute the **accuracy**, **precision**, **recall**, and **F_1 score**. Show your working. *(6 marks)*

(c) A colleague says "the classifier has 80% accuracy, so it is good." The positive class makes up only a small fraction of a much larger dataset. Explain why accuracy can be misleading here and which metrics you would report instead. *(4 marks)*

Question 2. (12 marks)

(a) Explain the **precision/recall trade-off** in terms of the classifier's **decision threshold**. What happens to precision and recall as the threshold is raised? *(4 marks)*

(b) For each scenario, state whether you would favour **high precision** or **high recall**, and justify: *(4 marks)* 1. A classifier that flags videos as "safe for children." 2. A classifier that screens medical scans for a dangerous tumour.

(c) Recall only ever decreases as the threshold is raised, but precision can sometimes *dip*. Explain why precision is not strictly monotonic in the threshold. *(4 marks)*

Question 3. (12 marks)

(a) Describe how the **ROC curve** is constructed: name the quantity on each axis and define it. *(4 marks)*

(b) What does the **AUC** measure? State the AUC of a perfect classifier and of a purely random one, and explain how to read the curve to judge classifier quality. *(4 marks)*

(c) A dataset is highly imbalanced (positives are rare). Explain why the **precision/recall curve** may be more informative than the ROC curve in this case. *(4 marks)*

Question 4. (12 marks)

Consider the email bag-of-words training set below. The dictionary order is **[Money, Free, Win, Offer, Meeting, Report]**.

i	Money	Free	Win	Offer	Meeting	Report	Spam
1	2	1	1	0	0	0	True
2	1	1	0	1	0	0	True
3	0	0	0	0	1	1	False
4	0	0	0	1	1	1	False

A new query email is **"free offer"**.

(a) Write the bag-of-words feature vector for the query. *(2 marks)*

(b) Using **Euclidean distance**, compute the distance from the query to each of the four training emails. Show your working. *(5 marks)*

(c) What label does a **k = 1** nearest-neighbour classifier predict? What does a **k = 3** classifier predict? Show the votes. *(3 marks)*

(d) Give **two** reasons why feature scaling matters for k-NN, and name the hyperparameter that controls the bias/variance trade-off in k-NN. *(2 marks)*

Question 5. (12 marks)

(a) Explain the difference between **multiclass** and **multilabel** classification. Give a real-world example of each, describing the features and the response. *(5 marks)*

(b) Explain the **OvR (one-versus-the-rest)** and **OvO (one-versus-one)** strategies. For N classes, how many binary classifiers does each require, and why is OvO often preferred for SVMs? *(4 marks)*

(c) A `RandomForestClassifier` does not expose a `decision_function()`. Write one line of code showing how you would obtain per-instance scores suitable for plotting a precision/recall curve, and explain what those scores represent. *(3 marks)*